



केन्द्रीय विद्युत अनुसंधान संस्थान

(भारत सरकार की सोसाईटी, विद्युत मंत्रालय)

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SHORT-CIRCUIT LABORATORY

SC/B/8/2018

Dt. 23rd August, 2018

TO WHOM-SO-EVER IT MAY CONCERN

This is to inform whom so ever it may concern that M/s. Balaji Electro Controls Pvt. Ltd., No. 36/2, Madanayakanahalli, Near Bhoruka School, Tumkur Road, Bengaluru - 562 123 have conducted following Design Verification Tests on LV Switchgear & controlgear Assembly - 415V (Ue) 3200A 3Ph+N 660V (Ui) 8 kV (Uimp) 50Hz IP42 Icw-50kA/1s Icc-50kA Form 4b construction Modular Panel comprising of 3200A TP ACB incomer, 1600A TP ACB, 630 A, 400 A & 200A 3P MCCB outgoing, as per IEC 61439-1: 2011 & IEC 61439-2: 2011, during March to June 2018 at CPRI, Bengaluru.

SL. No.	List of Design Verification tests	Sub-clause	Remarks
1	Degree of Protection of Enclosure - IP 42	10.3	Complies
2	Temperature-rise Limits	10.10	Within the limit
3	Dielectric Properties		
	Power frequency withstand voltage	10.9.2	Withstood
	Impulse withstand voltage - 8 kV	10.9.3	Withstood
4	Short Circuit Withstand Strength - (50kA/1s on main bus-bars & outgoing feeder bus-bars including the incoming devices & 30kA/1sec. in neutral bus-bar, conditional SC test on incoming & outgoing at 50kA)	10.11	Withstood
5	Protection against electric shock & integrity of protective circuits	10.5	
	Effective continuity between exposed conductive parts of the assembly and the protective circuit	10.5.2	Complies
	Verification of short circuit strength of Protective Circuit	10.5.3	Withstood
6	Clearance & Creepage Distance	10.4	Complies
7	Incorporation of switching devices & components	10.6	Complies
8	Internal electrical circuits & connections	10.7	Complies
9	Mechanical Impact (IK 05)	10.2.6	Complies
10	Marking	10.2.7	Complies
11	Properties of insulating materials	10.2.3	
	Thermal stability	10.2.3.1	Complies
	Resistance of insulating materials to abnormal heat and fire due to internal electric effects	10.2.3.2	Complies
12	Mechanical operation	10.13	Complies
13	Lifting	10.2.5	Complies
14	Resistance to corrosion	10.2.2	Complies

The final Test Certificate will be issued after submission of final drawings. This letter is issued to M/s. Balaji Electro Controls Pvt. Ltd., Bengaluru - 562 123, on their request to enable them to submit to their end customer.

Recd on 23.8.18
[Signature]

[Signature] 23/08/2018
(Swaraj Kumar Das)
Additional Director